Exercise 12: Basic Controller for the Lateral Motion

The following state space model of a generic aircraft (Class II) in horizontal cruise flight with $V_0 = 150\frac{m}{s}, h = 1000m$) is given:

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

a) Determine the control structure and the corresponding controller gains such that the Roll Mode of the aircraft fulfills level 1 handling qualities according to MIL-F-8785C.
b) Determine the handling qualities level according to MIL-F-8785C for the Dutch Roll.
c) Propose a controller structure that changes the relative damping of the Dutch Roll.
d) Determine the gain which can be used to achieve a general, desired relative damping. You can assume that the eigenfrequency of the Dutch Roll remains unchanged by the feedback control.